

4.8.2 Resources of materials and energy

The example of metal extraction is used to show how the Earth's natural resources are used to manufacture useful products. A variety of renewable and non-renewable energy resources is needed for these and other aspects of modern life. In order to operate sustainably, scientists and engineers seek to minimise the use of limited resources, cut energy consumption, reduce waste and limit environmental impacts. Scientists also aim to develop ways of disposing of products at the end of their useful life in ways that ensure that materials and stored energy are utilised. Life cycle assessments can be used to compare the overall impact of the production, use and disposal of products.

4.8.2.1 Metal extraction by reduction of oxides

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain reduction and oxidation in terms of loss or gain of oxygen, identifying which species are oxidised and which are reduced. Explain, using the position of carbon in the reactivity series, the principles of industrial processes used to extract metals, including extraction of a non-ferrous metal.	Metals react with oxygen to produce metal oxides. These are oxidation reactions. Reduction involves the loss of oxygen. Unreactive metals such as gold are found in the Earth as the metal itself but most metals are found as compounds that require chemical reactions to extract the metal. Knowledge and understanding are limited to the reduction of oxides using carbon. Metals less reactive than carbon can be extracted from their oxides by reduction with carbon. Knowledge of the details of processes used in the extraction of metals is not required.	WS 1.2 Identify the substances which are oxidised or reduced in terms of gain or loss of oxygen. WS 1.4 Explain in terms of the reactivity series why some metals are extracted with carbon and others by electrolysis. Interpret or evaluate specific metal extraction processes when given appropriate information.

4.8.2.2 Metal extraction by electrolysis

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain why and how electrolysis is used to extract some metals from their ores.	Metals can be extracted from molten compounds using electrolysis. Electrolysis is used if the metal is too reactive to be extracted by reduction with carbon or if the metal reacts with carbon. Large amounts of energy are used in the extraction process to melt the compounds and to produce the electrical current. Aluminium is manufactured by the electrolysis of a molten mixture of aluminium oxide and cryolite using positive electrodes (anodes) made of carbon. The anodes have to be replaced from time to time.	WS 1.4 Explain technological applications of science, including the use of cryolite for the extraction of aluminium and the need to replace the anodes.

4.8.2.3 Metal extraction by biological methods (HT only)

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Evaluate alternative biological methods of metal extraction (bacterial and phytoextraction).	Copper ores are becoming scarce and new ways of extracting copper from low-grade ores include phytomining and bioleaching. These methods avoid traditional mining methods of digging, moving and disposing of large amounts of rock. Phytomining uses plants to absorb metal compounds. The plants are harvested and then burned to produce ash that contains metal compounds. Bioleaching uses bacteria to produce leachate solutions that contain metal compounds. The metal compounds can be processed to obtain the metal. For example, copper can be obtained from solutions of copper compounds by displacement using scrap iron or by electrolysis.	WS 1.4 Evaluate environmental implications of the applications of science.

4.8.2.4 Energy resources

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the main energy resources available for use on Earth (including fossil fuels, nuclear fuel, biofuel, wind, hydroelectricity, the tides and the Sun); compare the ways in which they are used and distinguish between renewable and non-renewable resources.	Non-renewable resources of energy include: • coal • crude oil • natural gas • nuclear fuel. A renewable energy resource is one that is being (or can be) replenished as it is used. Examples include: • plants that provide biofuel • wind turbines • hydroelectricity • tidal barrages or undersea turbines • solar panels that produce electricity or heat water.	WS 1.4 Explain technological applications of science. Explain patterns and trends in given data about the use of energy resources. Evaluate the use of different energy resources, taking into account reliability, cost and impact on the environment. WS 4.4, MS 1c, 2c, 4a Interpret data with energy quantities given, using the prefixes kilo, mega, giga and tera.

4.8.2.5 Energy conservation and dissipation

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe, with examples, where there are energy transfers in a system and where there is no net change to the total energy of a closed system (qualitative only). Describe, with examples, how in all system changes, energy is dissipated so that it is stored in less useful ways.	Energy can be transferred usefully, stored or dissipated, but cannot be created or destroyed. Whenever there are energy transfers in a system only part of the energy is usefully transferred. The rest of the energy is dissipated so that it is stored in less useful ways. This energy is often described as being 'wasted'.	WS 3.3, MS 1a, 1c, 3c Make calculations of the energy changes associated with changes in a system, recalling or selecting the relevant equations for mechanical, electrical and thermal processes; thereby express in quantitative form and on a common scale the overall redistribution of energy in the system.

4.8.2.6 Preventing unwanted energy transfers

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain ways of reducing unwanted energy transfer, eg through lubrication and thermal insulation; describe the effects, on the rate of cooling of a building, of the thickness and thermal conductivity of its walls (qualitative only).	Unwanted energy transfers can be reduced in a number of ways, for example through: - lubrication – work done against the frictional forces acting on an object causes a rise in the temperature of the object and dissipates useful energy - the use of thermal insulation – the higher the thermal conductivity of a material the higher the rate of energy transfer by conduction across the material.	WS 1.4 Explain technological applications of science.

4.8.2.7 Energy efficiency

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Calculate energy efficiency for any energy transfer, (HT only) and describe ways to increase efficiency.	The energy efficiency for any energy transfer can be calculated using the equation: $\text{efficiency} = \frac{\text{useful output energy transfer}}{\text{total input energy transfer}}$	WS 3.3, MS 3c Recall and apply this equation. MS 1a, 1c, 3c Calculate or use efficiency values as a decimal or as a percentage.

4.8.2.8 Life cycle assessment

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the basic principles in carrying out a life cycle assessment of a material or product. Interpret data from a life cycle assessment of a material or product.	Life cycle assessments (LCAs) are carried out to assess the environmental impact of the materials used and the energy resources needed for products in each of these stages: • extracting and processing raw materials • manufacturing and packaging • use and operation during its lifetime • disposal at the end of its useful life including transport and distribution at each stage. The use of water, energy resources and materials, as well as the production of some wastes, can be fairly easily quantified. Allocating numerical values to pollutant effects is less straightforward and requires value judgements, so LCA is not a purely objective process. Selective or abbreviated LCAs can be devised to evaluate a product but these can be misused to reach pre-determined conclusions, eg in support of claims for advertising purposes.	WS 1.3, 1.4, 3.3, 3.5 Interpret data from LCAs of materials or products given appropriate information. MS 1a Recognise and use expressions in decimal form. MS 1d Make estimates of the results of simple calculations. WS 4.6, MS 2a Use an appropriate number of significant figures. WS 3.5, MS 4a Translate information between graphical and numeric form.

4.8.2.9 Recycling

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe a process where a material or product is recycled for a different use, and explain why this is viable.	Reuse and recycling of materials by end users cuts down the use of limited material resources. It can also cut the use of energy resources and the production of waste. Metals can be recycled by melting and recasting or reforming into different products. The amount of separation required for recycling depends on the metal and the properties required of the final product. For example, in steel making some scrap steel is added to the iron from a blast furnace to reduce the amount of iron that needs to be extracted from iron ore.	WS 1.4 Evaluate factors that affect decisions on recycling, given appropriate information.